

Submission PY1-FINAL-SUB01

Question PY1-FINAL-Q01

scratch: show step

1. range(6) gives 0 through 5
 2. keep even i: 0, 2, 4
 3. squares => [0, 4, 16]
- => FINAL: [0, 4, 16] (check units/interpretation)

Question PY1-FINAL-Q02

scratch: show step

1. counts = Counter(xs)
 2. for x in xs: if counts[x] == 1: return x
 3. return None; two linear passes $\Rightarrow O(n)$ time, $O(n)$ space
- => FINAL: scan counts, then scan in order; return first count==1 else None; $O(n)$ time, $O(n)$ space (check units/interpretation)

Question PY1-FINAL-Q03

scratch: show step

1. A single pass fixes only adjacent inversions encountered once.
 2. Repeat until no swaps (or $n-1$ passes), shrinking the suffix after each pass.
- => FINAL: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (check units/interpretation)

Question PY1-FINAL-Q04

scratch: show step

1. linear scan: up to 64 comparisons, $\Theta(n)$
 2. binary search: at most $\lceil \log_2(64) \rceil + 1$ comparisons, $\Theta(\log n)$
- => FINAL: linear $\Theta(n)$, up to 64; binary $\Theta(\log n)$, about 7 (check units/interpretation)

Question PY1-FINAL-Q05

scratch: show step

1. Removing shifts later elements left while the iterator advances.
 2. Build a filtered list or iterate over a copy, then replace the original.
- => FINAL: mutation shifts elements and skips checks; use a filtered copy or iterate over a copy (check units/interpretation)

Question PY1-FINAL-Q06

scratch: show step

1. range(5) gives 0 through 4
 2. keep even i: 0, 2, 4
 3. squares => [0, 4, 16]
- => FINAL: [0, 4, 16] (check units/interpretation)

Question PY1-FINAL-Q07

scratch: show step

1. counts = Counter(xs)
 2. for x in xs: if counts[x] == 1: return x
 3. return None; two linear passes $\Rightarrow O(n)$ time, $O(n)$ space
- $\Rightarrow$ FINAL: scan counts, then scan in order; return first count==1 else None; $O(n)$ time, $O(n)$ space (check units/interpretation)

Question PY1-FINAL-Q08

scratch: show step

1. A single pass fixes only adjacent inversions encountered once.
 2. Repeat until no swaps (or $n-1$ passes), shrinking the suffix after each pass.
- $\Rightarrow$ FINAL: repeat passes or track the shrinking unsorted suffix; one pass alone is not a full sort (check units/interpretation)

Question PY1-FINAL-Q09

scratch: show step

1. linear scan: up to 64 comparisons, $\Theta(n)$
 2. binary search: at most $\lceil \log_2(64) \rceil + 1$ comparisons, $\Theta(\log n)$
- $\Rightarrow$ FINAL: linear $\Theta(n)$, up to 64; binary $\Theta(\log n)$, about 7 (check units/interpretation)

Question PY1-FINAL-Q10

scratch: show step

1. Removing shifts later elements left while the iterator advances.
 2. Build a filtered list or iterate over a copy, then replace the original.
- $\Rightarrow$ FINAL: mutation shifts elements and skips checks; use a filtered copy or iterate over a copy (check units/interpretation)